

Assignment No- 03

Subject- Operating System

1	What is a deadlock? Explain necessary conditions for deadlock.
2	Explain deadlock prevention.
3	Explain Banker's algorithm with suitable examples.
4	Explain the concept of resource allocation graph.
5	Explain deadlock avoidance.
6	Explain the concept of a safe state with suitable examples.
7	What are critical section problems and what are the possible solutions to the problems.
8	Explain the readers and writers problem with the help of semaphore.
9	Explain software approaches to solve synchronization problems.
10	Explain hardware approaches to solve synchronization problems.
11	Define process synchronization and explain Peterson's solution algorithm.
12	What is semaphore? Differentiate between mutex and semaphore.
13	Explain the producer and consumer problem with the help of semaphore.
14	Explain deadlock detection and recovery.
15	What is a monitor? Explain with any suitable example.
16	What is deadlock, and what are the necessary conditions for a deadlock to occur?
17	How can deadlock be prevented in a system?
18	Describe the Banker's algorithm with a relevant example.
19	What is a resource allocation graph, and how does it relate to deadlock detection?
20	How does deadlock avoidance work in a system?
21	What is the concept of a safe state, and can you explain it with an example?
22	What are critical section problems, and what solutions can be used to address these problems?
23	Explain the readers-writers problem and how semaphores can be used to solve it.
24	What are the software approaches for solving synchronization problems?
25	What are the hardware approaches for solving synchronization problems?
26	What is process synchronization, and how does Peterson's solution algorithm address it?
27	Define semaphore and explain the difference between a semaphore and a mutex.
28	Can you describe the producer-consumer problem and how semaphores help in solving it?
29	How does deadlock detection work, and what are the possible recovery methods?
30	What is a monitor in operating systems? Provide an example to illustrate its use.
31	What is a Resource Allocation Graph? Explain its significance in the context of deadlock detection.
32	What is deadlock avoidance, and how is it different from deadlock prevention?
33	Define a safe state in a system and explain it with a suitable example.
34	What are critical section problems in process synchronization, and what are the potential solutions to these problems?
35	Explain the Readers-Writers problem and how semaphores can be used to solve it.
36	What are the software-based approaches to solve synchronization problems in operating systems?
37	Describe the hardware-based approaches to solving synchronization issues.
38	What is process synchronization? Explain Peterson's solution algorithm in the context of process synchronization.
39	What is a semaphore? Differentiate between a mutex and a semaphore.
40	Explain the Producer-Consumer problem using semaphores as a solution.